INCH-POUND

MIL-DTL-60561C
14 December 2010
SUPERSEDING
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DETAIL SPECIFICATION

THRUSTER, CARTRIDGE ACTUATED, M25A1 METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M25A1 Cartridge Actuated Thruster Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-2550 Ammunition, General Specification For

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-413 Visual Inspection Guide for Elastomeric O-rings

(Copies of these documents are available online at https://assist.daps.dla.mil/ or from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia PA 19111- 5094.)

Comments, suggestions, or questions on this document should be addressed to Department of the Navy, Indian Head Division, NSWC, Code E12P, Document Control, 4123 Artisans Court, Suite 103, Indian Head, Maryland 20640-5085, OFFICIAL BUSINESS, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil/.

AMSC N/A FSC 1377

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2.2.2 Other Government documents, drawings, and publications. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

11730535	Thruster, Cartridge Actuated, M25A1 Metal Parts Assembly
8797741	Box, Fiberboard (Inner), Packing, Ammunition, For Thruster, Cartridge Actuated, M5A2, M20A1, or M25A1
8797742	Box, Packing, Ammunition, For Thruster, Cartridge Actuated, M5A2, M20A1, or M25A1
9217579	Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Thruster, Cartridge Actuated, M5A2, M20A1, or M25A1

U.S. Air Force

TPO 00-554-3103	Thruster, Cartridge Actuated, M25A1

(Application for copies should be addressed to the DEPARTMENT OF THE NAVY, Indian Head Division, NSWC, Code E12P, Document Control, 4123 Artisans Court, Suite 103, Indian Head, Maryland 20640-5085, OFFICIAL BUSINESS.)

2.3 Non-Government publications and drawings. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)/AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ANSI/ASQ Z1.4	Sampling Procedures And Tables For Inspection By Attributes
ANSI/ISO/ASQ A8402	Quality Assurance Terms And Definitions

(Copies of this document are available online at http://e-standards.asq.org/ or from the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to the first article inspection (see 4.4 and 6.3).

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3.2 Materials. All materials used in the manufacture of the thrusters shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Raw material. The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

3.3 Hardness and other physical properties. Hardness and other physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.4 Hydrostatic pressure. The body and head shall withstand a hydrostatic pressure of 8,000 pounds per square inch (psi) for 15 seconds, minimum, without leaking, deformation, or mechanical failure (see 6.4.1).

3.5 Structural strength. The assembled thruster shall withstand a static tensile force of 2,000 pounds, minimum, for 15 seconds, minimum, between mounting points without unlocking and without mechanical failure (see 6.4.1).

3.6 Shear pin strength. The shear pin, after application of protective finish, shall shear when a load of 48 ± 4 pounds is applied at the steady rate of 0.125 inch per minute.

3.7 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled handtight.

3.8 Protective finish. Protective finish shall be in accordance with the applicable specifications and shall apply to all surfaces of the component regardless of contour configuration.

3.9 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using procedures in 4.6.6.

3.10 Workmanship. Thrusters shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.10.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.10.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.10.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. VERIFICATION

4.1 Classification of inspections.

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a.First article inspection (see 4.4)
b.Quality conformance inspection (see 4.5)

4.1.1 Quality assurance terms and definitions. Reference shall be made to ANSI/ISO/ASQ A8402 to define the quality assurance terms used.

4.2 Inspections. Inspections shall be in accordance with MIL-A-2550.

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a.Temperature: room ambient (68 to 77°F)
b.Altitude: sea level
c.Humidity: 50 percent relative, maximum
d.Atmospheric pressure: 14.7 psi

4.4 First article inspection. At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of five metal parts assemblies, nine sets of unassembled parts, one set of unassembled parts without plating or final protective finish, and fifty shear pins from the same lot of shear pins tested by the contractor. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the first article sample shall be inspected for all requirements of the drawings and specifications. The Government reserves the right to require new first article samples until an acceptable sample is submitted (see 6.3).

4.5 Quality conformance inspection.

4.5.1 Lot.

4.5.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with ANSI/ASQ Z1.4.

4.5.1.2 Size of lot. The production lot size shall be as specified in the contract.

4.5.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

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4.5.2.1 Key, Drawing 10534180, covering a detail of Drawing 11730535.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Surface finish incorrect	Visual
102	Protective finish inadequate or defective	Visual
103	Metal defective	Visual
104	Burr or sharp corner	Visual
105	Foreign matter	Visual
Minor		
201	Total length	Gage
202	Length from end to vertical center	Gage
203	Length from top to true position of longitudinal center	Gage
204	Length from bottom of true position of longitudinal center	Gage
205	Width	Gage
206	Height	Gage
207	Radius or chamfer incorrect	Gage
208	Radius or chamfer missing	Visual

4.5.2.2 Pin, Firing, Drawing 8593864, covering a detail of Drawing 11730535.

Critical		
None defined		
Major		
101	Concentricity of groove with body	Gage
102	Concentricity of striker with body	Gage
103	Concentricity of striker shoulder with body	Gage
104	True position of hole	Gage
105	Diameter of groove	Gage
106	Diameter of body, min	Gage
107	Diameter of striker, max	Gage
108	Diameter of hole, max	Gage
109	Profile of striker	Gage
110	Profile of groove	Gage
111	Metal defective	Visual
112	Protective finish inadequate or defective	Visual
113	Surface finish of body and striker tip incorrect	Visual
114	Burr or sharp corner	Visual
115	Foreign matter	Visual
Minor		
201	Diameter of body, max	Gage
202	Diameter of striker, min	Gage

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203	Diameter of hole, min	Gage
204	Diameter of striker shoulder	Gage
205	Depth of hole	Gage
206	Length from body to tip of striker	Gage
207	Length of striker	Gage
208	Total length	Gage
209	Location of hole	Gage
210	Location of groove	Gage
211	Depth of slot in base	Gage
212	Length of body diameter forward of groove, min	Gage
213	Width of groove	Gage
214	Width of slot in base	Gage
215	Location of slot in base	Gage
216	Radius incorrect	Gage
217	Radius missing	Visual
218	Surface finish improper, except as otherwise classified	Visual
219	Marking missing, incorrect, or illegible	Visual

4.5.2.3 Body Assembly, Drawing 10534179, covering a detail of Drawing 11730535.

Critical
None defined

Major

101	Pitch diameter of thread, min	Gage
102	Major diameter of thread, min	Gage
103	Concentricity of pitch diameter of thread with cavity	Gage
104	Concentricity of bore with cavity	Gage
105	Concentricity of groove with bore	Gage
106	Concentricity of small counterbore with bore	Gage
107	Concentricity of large counterbore with bore	Gage
108	True position of slots	Gage
109	Symmetry of slots with each other and body datum diameter	Gage
110	Diameter of thread undercut, min	Gage
111	Diameter of cavity, max	Gage
112	Diameter of internal undercut, max (if applicable)	Gage
113	Diameter of bore	Gage
114	Diameter of groove	Gage
115	Diameter of small counterbore	Gage
116	Diameter of body at threaded end	Gage
117	Location of groove	Gage
118	Deleted	
119	Straightness of cavity	Gage
120	Width of groove	Gage
121	Profile of groove	Gage
122	Thread damaged or not full	Visual
123	Metal defective	Visual
124	Protective finish inadequate or defective	Visual
125	Surface finish of cavity improper	Visual

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126	Burr or sharp corner	Visual
127	Foreign matter	Visual
128	Deleted	
Minor		
201	Pitch diameter of thread, max	Gage
202	Diameter of cavity, min	Gage
203	Diameter of large counterbore	Gage
204	Diameter of body, max	Gage
205	Width of flat at threaded face	Gage
206	Thickness of wall	Gage
207	Length from end to shoulder of large counterbore	Gage
208	Length from end to large shoulder of small counterbore	Gage
209	Total length	Gage
210	Width across flats	Gage
211	Depth of slots	Gage
212	Width of slots	Gage
213	Deleted	
214	Length from front tapered face to exterior shoulder	Gage
215	Deleted	
216	Deleted	
217	Deleted	
218	Width of thread undercut, max	Gage
219	Width of interior undercut, max	Gage
220	Length from end to inside front face of bore	Gage
221	Diameter of exterior shoulder	Gage
222	Width of exterior shoulder	Gage
223	Location of undercut from shoulder	Gage
224	Location of flats incorrect	Gage
225	Deleted	
226	Radius or chamfer incorrect	Gage
227	Radius or chamfer missing	Visual
228	Surface finish improper, except as otherwise specified	Visual
229	Marking missing, incorrect, or illegible	Visual

4.5.2.4 Trunnion, Drawing 8594482, covering a detail of Drawing 11730535.

Critical
None defined

Major		
101	Concentricity of lugs	Gage
102	True position of lugs	Gage
103	Symmetry of flats at root of lugs with inside diameter	Gage
104	Perpendicularity of sides with inside diameter	Gage
105	Diameter of lugs, min	Gage
106	Width across flats, min	Gage
107	Radius at base of lugs	Gage

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108	Metal defective	Visual
109	Protective finish inadequate or defective	Visual
Minor		
201	Diameter of lugs, max	Gage
202	Outside diameter, max	Gage
203	Inside diameter	Gage
204	Length of lugs	Gage
205	Width across flats, max	Gage
206	Width of body	Gage
207	Thickness of wall	Gage
208	Location of lugs	Gage
209	Radius or chamfer incorrect, except as otherwise classified	Gage
210	Radius or chamfer missing	Visual
211	Surface finish improper	Visual
212	Burr or sharp corner	Visual
213	Foreign matter	Visual
214	Marking missing, incorrect, or illegible	Visual

4.5.2.5 Spring, Piston Locking, Drawing 10534184, covering a detail of Drawing 11730535.

Critical
None defined

Major		
101	Load at specified height	Gage
102	Solid length, max	Gage
103	Ends incorrect	Visual
104	Protective finish inadequate or defective	Visual
105	Burr	Visual

Minor
None defined

4.5.2.6 Head, Drawing 10521455, covering a detail of Drawing 11730535.

Critical
None defined

Major		
101	Pitch diameter of small thread, max	Gage
102	Minor diameter of small thread, max	Gage
103	Pitch diameter of midsection thread, max	Gage
104	Minor diameter of midsection thread, max	Gage
105	Pitch diameter of large thread, max	Gage
106	Minor diameter of large thread, max	Gage
107	Concentricity of pitch diameter of large thread with pitch diameter of midsection thread	Gage
108	Depth of cavity, small end	Gage

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109	Depth of counterbore, small end	Gage
110	Length of thread, small end, max	Gage
111	Length of effective thread, small end, min	Gage
112	Diameter of undercut, large thread, max	Gage
113	Diameter of undercut, midsection thread, max	Gage
114	Diameter of chamfer at small end	Gage
115	Angle of chamfer at small end	Gage
116	Thread damaged or not full	Visual
117	Metal defective	Visual
118	Protective finish inadequate or defective	Visual
119	Surface finish of face in midsection cavity, chamfers in small end and small thread face improper	Visual
120	Burr	Visual
121	Foreign matter	Visual

Minor

201	Pitch diameter of small thread, min	Gage
202	Pitch diameter of midsection thread, min	Gage
203	Pitch diameter of large thread, min	Gage
204	Diameter of counterbore, small end	Gage
205	Diameter of hole through midsection	Gage
206	Depth of large cavity	Gage
207	Diameter of body, midsection	Gage
208	Diameter of body, large end	Gage
209	Diameter of body, small end	Gage
210	Depth to bottom of midsection	Gage
211	Length of large body diameter	Gage
212	Length of small body diameter	Gage
213	Length of flat	Gage
214	Total length	Gage
215	Width of undercut, large thread	Gage
216	Width of undercut, midsection thread	Gage
217	Width across flats	Gage
218	Location of flats	Gage
219	Radius or chamfer incorrect, except as otherwise classified	Gage
220	Radius or chamfer missing	Visual
221	Surface finish improper, except as otherwise classified	Visual
222	Marking missing, incorrect, or illegible	Visual

4.5.2.7 Piston, Drawing 10534181, covering a detail of Drawing 11730535.

Critical
None defined

Major

101	Pitch diameter of thread, min	Gage
102	Major diameter of thread, min	Gage
103	Pitch diameter of small threads, max	Gage
104	Minor diameter of small threads, max	Gage

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105	Concentricity of small thread holes minor diameter	Gage
106	Concentricity of body midsection with large end	Gage
107	Concentricity of shoulder with large end	Gage
108	Concentricity of groove with body, large end	Gage
109	Concentricity of large cavity with large outside shoulder	Gage
110	Concentricity of small cavity with large cavity	Gage
111	Concentricity of intermediate cavity with large cavity	Gage
112	Symmetry of slots	Gage
113	True position of holes with body	Gage
114	Diameter of groove	Gage
115	Diameter of body at large end, min	Gage
116	Diameter of body between wrench flats and undercut, min	Gage
117	Diameter of large cavity	Gage
118	Location of slots	Gage
119	Diameter of intermediate cavity	Gage
120	Location of large shoulder of intermediate cavity	Gage
121	Profile of groove	Gage
122	Thread damaged or not full	Visual
123	Metal defective	Visual
124	Protective finish inadequate or defective	Visual
125	Surface finish improper	Visual
126	Burr or sharp corner	Visual
127	Foreign matter	Visual
Minor		
201	Pitch diameter of thread, max	Gage
202	Diameter of body at large end, max	Gage
203	Diameter of shoulder, large end	Gage
204	Pitch diameter of small thread, min	Gage
205	Diameter of drill point	Gage
206	Diameter of undercut, min (if applicable)	Gage
207	Diameter of small cavity	Gage
208	Length of body at large end	Gage
209	Depth of holes	Gage
210	Length of slots	Gage
211	Width of slots	Gage
212	Profile of slots	Gage
213	Length of wrench flats	Gage
214	Length of shoulder, large end	Gage
215	Location of holes	Gage
216	Diameter of large cavity, min	Gage
217	Length from shoulder to small end of body	Gage
218	Depth of large cavity	Gage
219	Total length	Gage
220	Diameter of face	Gage
221	Width of undercut, max (if applicable)	Gage
222	Location of groove	Gage
223	Width across wrench flats	Gage
224	Length of thread	Gage

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225	Radius or chamfer incorrect	Gage
226	Radius or chamfer missing	Visual
227	Surface finish improper, except as otherwise classified	Visual
228	Marking missing, incorrect, or illegible	Visual
229	Diameter (2) holes through wrench flats	Gage
230	Location (2) holes through wrench flats	Gage

4.5.2.8 Guide Retainer Assembly, Drawing 10521616, covering a detail of Drawing 11730535.

Critical
None defined

Major

101	Pitch diameter of external thread, min	Gage
102	Major diameter of external thread, min	Gage
103	Pitch diameter of internal thread, max	Gage
104	Minor diameter of internal thread, max	Gage
105	Concentricity of large cavity with pitch diameter of internal thread	Gage
106	Concentricity of boss with pitch diameter of internal thread	Gage
107	Concentricity of body with pitch diameter of external thread	Gage
108	Concentricity of small cavity with pitch diameter of internal thread	Gage
109	Concentricity of groove with body	Gage
110	Concentricity of hole through boss with pitch diameter of internal thread and small cavity	Gage
111	Concentricity of recess with small cavity	Gage
112	Perpendicularity of boss face and base of small cavity with pitch diameter of internal thread	Gage
113	True position of hole	Gage
114	Diameter of groove	Gage
115	Diameter of small cavity	Gage
116	Diameter of large cavity, max	Gage
117	Diameter of hole through boss	Gage
118	Diameter of body	Gage
119	Diameter of undercut, max	Gage
120	Height of boss	Gage
121	Thickness through boss	Gage
122	Thickness of base of small cavity	Gage
123	Deleted	
124	Profile of groove	Gage
125	Surface finish in small cavity improper	Visual
126	Metal defective	Visual
127	Protective finish inadequate or defective	Visual
128	Thread damaged or not full	Visual
129	Deleted	
130	Burr or sharp corner	Visual
131	Foreign matter	Visual

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Minor

201	Pitch diameter of external thread, max	Gage
202	Pitch diameter of internal thread, min	Gage
203	Symmetry of flats with pitch diameter of internal thread	Gage
204	Diameter of recess at base of small cavity	Gage
205	Diameter of large cavity, min	Gage
206	Diameter at rear of external thread	Gage
207	Diameter of pin hole	Gage
208	Deleted	
209	Depth of large cavity	Gage
210	Location of flange	Gage
211	Total length	Gage
212	Width across flats	Gage
213	Location of hole	Gage
214	Location of groove	Gage
215	Deleted	
216	Deleted	
217	Location of undercut	Gage
218	Deleted	
219	Width of groove	Gage
220	Length of minimum effective groove	Gage
221	Width of undercut at internal thread	Gage
222	Length of flats	Gage
223	Radius or chamfer incorrect	Gage
224	Surface finish improper, except as otherwise classified	Visual
225	Marking missing, incorrect, or illegible	Visual

4.5.2.9 Sleeve Assembly, Drawing 10521617, covering a detail of Drawing 11730535.

Critical
None defined

Major

101	Pitch diameter of thread, min	Gage
102	Major diameter of thread, min	Gage
103	Concentricity of bore with pitch diameter of thread	Gage
104	Diameter of bore	Gage
105	Deleted	
106	Deleted	
107	Deleted	
108	Deleted	
109	Deleted	
110	Radius at end of bore improper	Visual
111	Thread damaged or not full	Visual
112	Metal defective	Visual
113	Protective finish inadequate or defective	Visual
114	Surface finish of radius at end of bore improper	Visual
115	Burr	Visual
116	Foreign matter	Visual

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Minor

201	Pitch diameter of thread, max	Gage
202	Symmetry of flats with pitch diameter of thread	Gage
203	Width across flats	Gage
204	Total length	Gage
205	Length of thread	Gage
206	Length of flats	Gage
207	Radius or chamfer incorrect	Gage
208	Radius or chamfer missing, except as otherwise classified	Visual
209	Surface finish improper, except as otherwise classified	Visual
210	Marking missing, incorrect, or illegible	Visual

4.5.2.10 Pin, Shear, Drawing 10521618, covering a detail of Drawing 11730535.

Critical
None defined

Major

101	Diameter, max	Gage
102	Metal defective	Visual
103	Protective finish inadequate or defective	Visual
104	Burr	Visual
105	Foreign matter	Visual

Minor

201	Total length	Gage
202	Profile of end	Visual

4.5.2.11 Piston, Unlocking, Drawing 10534182, covering a detail of Drawing 11730535.

Critical
None defined

Major

101	Concentricity of groove with large outside diameter	Gage
102	Concentricity of body with large outside diameter	Gage
103	Concentricity of cavity with large outside diameter	Gage
104	Concentricity of small diameter at cavity end with large outside diameter	Gage
105	Concentricity of immediate diameter at cavity end with large outside diameter	Gage
106	Concentricity of large diameter at cavity end with large outside diameter	Gage
107	Diameter at large end, min	Gage
108	Diameter of groove, min	Gage
109	Length from face at cavity end to taper, max	Gage
110	Depth of cavity	Gage
111	Width of groove, max	Gage
112	Profile of groove	Gage

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113	Chamfer improper or incorrect	Visual
114	Protective finish inadequate or defective	Visual
115	Burr or sharp corner	Visual
116	Foreign matter	Visual

Minor

201	Diameter of large end, max	Gage
202	Diameter of groove, max	Gage
203	Diameter of body	Gage
204	Diameter of cavity	Gage
205	Small diameter at cavity end	Gage
206	Immediate diameter at cavity end	Gage
207	Large diameter at cavity end	Gage
208	Diameter of drill point, max	Gage
209	Angle at tapered surface	Gage
210	Width of groove, min	Gage
211	Thickness of lip	Gage
212	Length from face at cavity end to taper, min	Gage
213	Total length	Gage
214	Length from front face to body	Gage
215	Location of groove	Gage
216	Length from face of cavity end to body	Gage
217	Radius or chamfer missing, except as specified	Visual
218	Marking missing, incorrect, or illegible	Visual
219	Surface finish improper	Visual

4.5.2.12 Ring, Drawing 10534183, covering a detail of Drawing 11730535.

Critical
None defined

Major

101	Concentricity of exterior diameter with internal diameter	Gage
102	Concentricity of small holes	Gage
103	Concentricity of center bore with small holes	Gage
104	Chamfers at exterior and interior surfaces	Visual
105	Protective finish inadequate or defective	Visual
106	Burr or sharp corner	Visual
107	Foreign matter	Visual

Minor

201	Diameter at exterior surface	Gage
202	Diameter at interior surface	Gage
203	Diameter of holes	Gage
204	Diameter of counterbore	Gage
205	Depth of counterbore	Gage
206	Length from front face to rear face	Gage
207	Location of holes	Gage
208	Radius or chamfer incorrect	Gage

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209	Radius or chamfer missing	Visual
210	Surface finish improper	Visual
211	Marking missing, incorrect, or illegible	Visual

4.5.2.13 Assembly, Drawing 11730535, less all components except 10521616, 8593864, and MS29513-12.

Critical

1	Firing pin protrusion	Gage

Major

101	Any part missing	Visual
102	Protective finish inadequate or defective	Visual
103	Burr	Visual
104	Foreign matter	Visual

Minor
None defined

4.5.2.14 Assembly, Drawing 11730535.

Critical
None defined

Major

101	Trunnion not free to rotate 360 degrees	Manual
102	Any part missing, except as otherwise classified	Visual
103	Improper assembly	Visual
104	Thread damaged or not full	Visual
105	Marking missing,incorrect, or illegible	Visual
106	Burr	Visual
107	Foreign matter	Visual

Minor

201	Protective cap or plug missing	Visual
202	Protective finish inadequate or defective	Visual

4.5.2.15 Box, Fiberboard (Inner), Packing, Ammunition, Drawing 8797741.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual
202	Box damaged	Visual
203	Marking missing, incorrect, or illegible	Visual

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204	Cut, damaged, or improperly sealed bag	Visual

4.5.2.16 Box, Packing, Ammunition, Drawing 8797742.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Visual/Manual
202	Strapping missing, loose, broken, or improperly assembled	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.5.2.17 Box, Fiberboard (Outer), Alternative, Packing, Ammunition, Drawing 9217579.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual/Visual
202	Carton broken or damaged	Visual
203	Number of units incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.5.3 Inspection testing. The following tests shall be performed in accordance with 4.6.

4.5.3.1 Hardness. Fifty units from each heat treated lot of components shall be tested for compliance with the requirements on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.5.3.2 Hydrostatic pressure. Each body and head shall be subjected to the hydrostatic pressure test. Failure of any unit to comply with 3.4 shall be cause for rejection of the subassembly.

4.5.3.3 Structural strength. Each thruster shall be subjected to the structural strength test. Failure of any thruster to comply with the requirements of 3.5 shall be cause for rejection of the thruster.

4.5.3.4 Shear pin strength. Fifty pins from each lot of shear pins shall be tested for shear strength. Failure of any shear pin to comply with 3.6 shall be cause for rejection of the lot.

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4.5.3.5 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.5.3.6 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.6.6. Failure of any unit in the inspection sample to comply with 3.9 shall be cause for rejection of the assembly.

4.6 Test methods and procedures.

4.6.1 Hardness. Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the requirements of the applicable drawing.

4.6.2 Hydrostatic pressure. Prior to application of protective finish the body and end cap shall be mounted in the device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with 3.4.

4.6.3 Structural strength. The thruster shall be subjected to the structural strength test in accordance with the requirements of 3.5.

4.6.4 Shear pin strength. After application of protective finish, the pin shall be subjected to the shear pin strength test to determine compliance with 3.6. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.6.5 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.6.6 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.7 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point’s packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency’s automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The thruster assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a.Title, number, and date of this specification

b.If required, the specific issue of individual documents referenced (see 2.2.1 and 2.2.2)

c.If first article inspection is required (see 3.1 and 6.3)

d.The responsible inspection activity (see 4.1)

e.Any special instruction requirements (see 4.3)

f.Level of packing and marking (see 5.1)

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors that the items should be a first article sample. The first article should consist of 5 metal parts assemblies, 9 sets of unassembled parts, 1 set of unassembled parts without plating or final protective finish, and 50 shear pins. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definition.

6.4.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5 Packaging

6.5.1 Packing and marking. Packing and marking should be Level A or B, as specified by the contracting activity (see 6.2). Level A should be in accordance with Drawings 8797741 and 8797742. Level B should be in accordance with Drawings 8797741 and 9217579.

6.5.2 Air Force packaging. Air Force packaging should be in accordance with TPO 00-554-3103.

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6.6 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.7 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes. The changes in this issue were authorized by ECP 09E215A038DB.

Custodian:
Navy - OS
Air Force - 99

Preparing activity
Navy - OS
(Project1377-2010-005)

Review activity:
Air Force – 70

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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